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Artificial Intelligence for Network Traffic Management

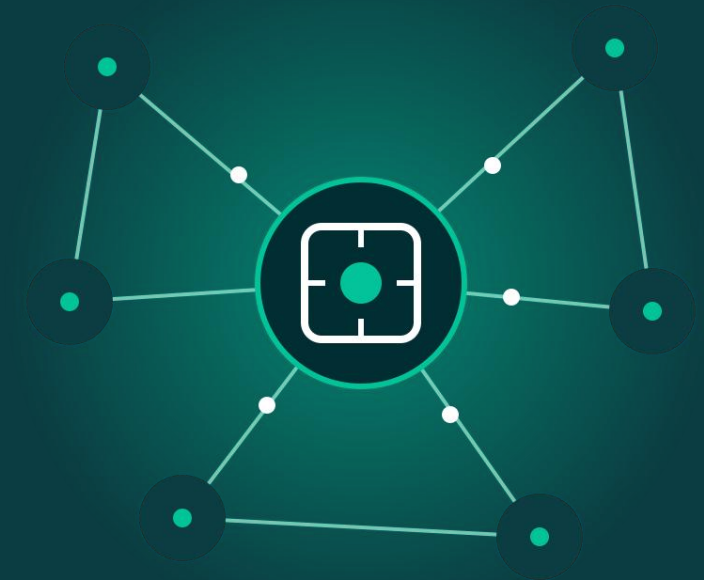
CSA0707-COMPUTER NETWORK

Under the Guidance Of :

Dr . R . Shankar

Presented By :

Avinash M (192521196)





Introduction



AI monitors network traffic in real time, watching how data moves across the system.



It studies patterns to predict congestion before it actually happens.



It automatically adjusts routing and bandwidth to keep traffic balanced.



It detects unusual behavior that could signal an attack or fault.



It replaces slow, manual, rule-based network control with smart automation.

Key AI Techniques Used



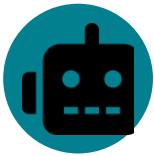
Machine Learning: learns traffic patterns from historical data.



Deep Learning: detects complex patterns across large networks.



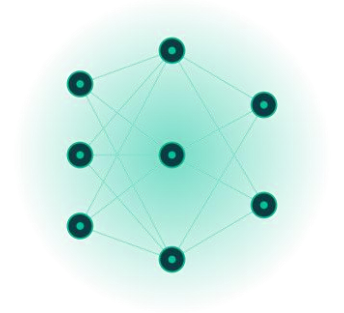
Predictive Analytics: forecasts future traffic spikes and slowdowns.



Reinforcement Learning: learns the best routing decisions by trial and reward.



Anomaly Detection: flags traffic that looks abnormal or suspicious.



Benefits



Faster Speeds: reduces delay by routing data through the best paths.



Better Efficiency: uses bandwidth more effectively, avoiding waste.



Stronger Security: Attacks earlier than manual checks.



Less Manual Work: automates decisions engineers once made by hand.



Scalability: handles growing traffic without breaking down.



Advantages

- Detects cyber threats in real time.
- Improves the accuracy of threat detection.
- Reduces human effort through automation.
- Responds quickly to security incidents.
- Protects networks from malware

Disadvantages

- Requires large amounts of quality data
- High implementation and maintenance cost.
- Can produce false positives and false negatives.
- Cybercriminals can use AI to create advanced attacks.
- Needs regular updates

Real-World Applications

Where AI traffic management is already at work today



Internet Service Providers: manage traffic smoothly for millions of users.



Data Centers: AI balances server loads and prevents equipment overload.



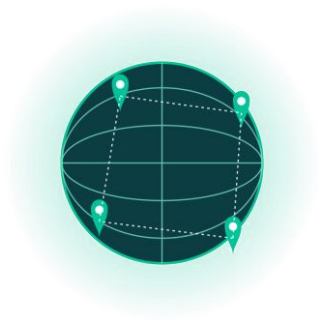
Video Streaming: platforms use AI to cut buffering during peak hours.



Smart Cities: connected devices and sensors are managed with AI traffic tools.



Cloud Computing: providers dynamically allocate bandwidth across services.



Challenges



AI needs large amounts of quality traffic data to learn well.



Deploying AI systems can be costly and technically complex.



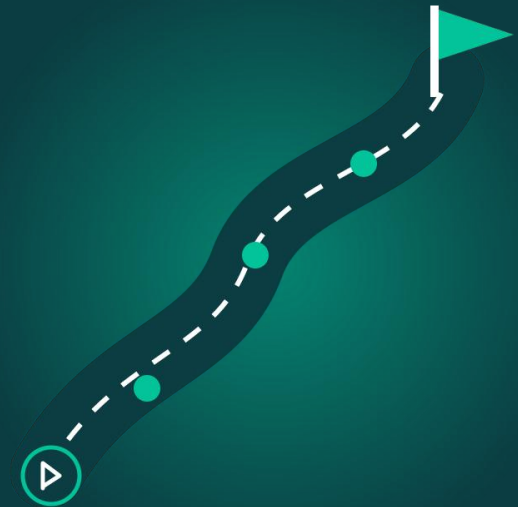
Poorly trained models may make wrong or biased decisions.



Networks must stay adaptable as traffic patterns keep evolving.

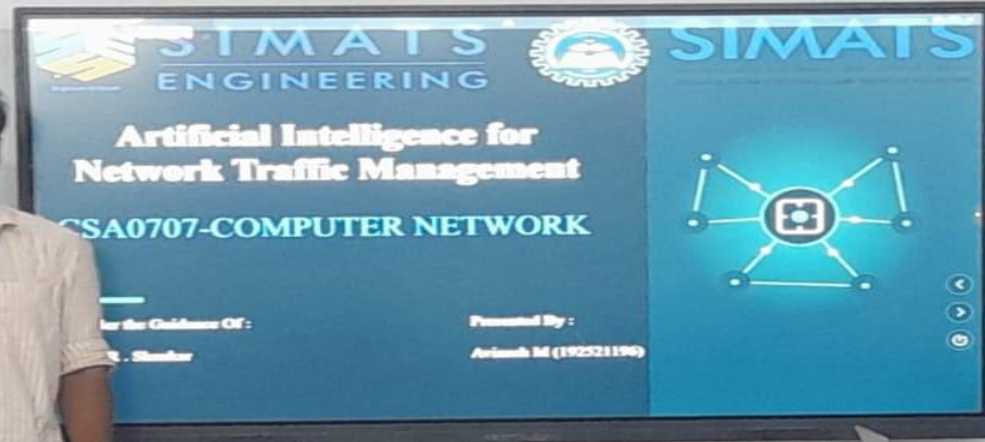


The future points to fully self-managing, self-healing networks.

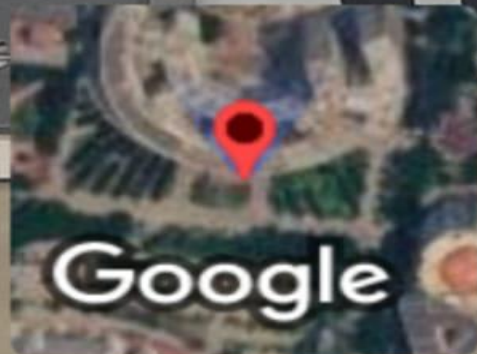


Conclusion

- AI improves network performance by managing traffic intelligently and efficiently.
- It predicts network congestion and reduces delays using real-time analysis.
- AI enhances network security by detecting unusual traffic and cyber threats.
- Automated traffic management reduces manual effort and improves resource utilization.
- AI-powered networking provides a scalable, reliable, and future-ready solution for modern communication systems.



GPS Map Camera



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THANK YOU...